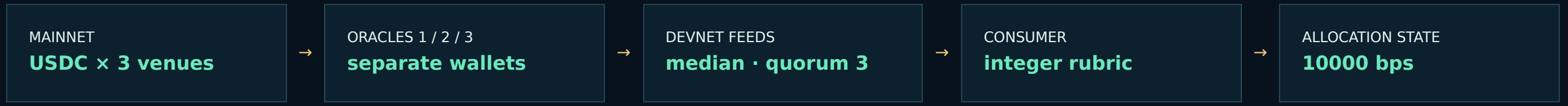
Three oracles. One allocation.

Live USDC supply APY from Marginfi, Save and Kamino, published by three independent signing processes.

3 / 3	4	203	4
sample quorum	observed allocations	fetched transactions	running containers

2026-10-10T12:27:31.708Z → 2026-10-10T12:36:41.151Z · mainnet reads / devnet writes

From protocol state to allocation



All three oracles independently read all three protocols. Each signs one vote per venue and devnet slot window. The consumer reads finalized feed heads from chain.

> 240-slot sample windows. One consumer ⇒ allocation quorum 1. Agreement is proven; source independence and consumer decentralization are outside this demo.

Deploy and verify the main program

DZGzK8wkxRWyhEorA8SSfPbUvHRAwog8vSsKeLEx4Coz

The existing devnet program was upgraded. Deployed bytecode was compared with the local SBF build, and upgrade authority was checked against the funding wallet.

slot 509534320 deploy / upgrade main program

3AZWKwDDCbDURRiQcSPCch2kfW3qkHDpxWPWDtrYkKN2djkWmJoECymiwh7ofgKuU9Me3hFDQWdnJ7ER6BaSmvC6

The registry and AllocationState already existed and were reused after identity and authority checks.

One funding wallet → three producers

Role	Public key	Funding
oracle-1	6mTM9YQcxGdkPmAzGCF25vCkdCpJqbjEYHxChvHfeTwH	0.25 devnet SOL
oracle-2	7tx3XxRt1u1bfsiwrjpaJ59hbr483Fvw1YiRmvs7ohZf	0.25 devnet SOL
oracle-3	FLj3KsBGXCh9uM7PY3ttKz13MpLvrj98i2qWY8daaa6L	0.25 devnet SOL

Funding wallet: `2rH2YbtCjFxNV4JL6JC1AQixxxviBEmMk8aBYBABoukc`

`slot 509535891` fund oracle-1 from existing devnet wallet

`5j8VkZ14dKg8xZEVpO7O4smW4na9RBO9nnNqQcmkHZEfD32CYkLSCNLP4eXi1TbVjneXAz6vL7R123BezBb2zh1k`

`slot 509535900` fund oracle-2 from existing devnet wallet

`4X6sdupzCYrPJiaYoeNygmR4XRi4uUD9B9RRj32xGqChWhw6yJ8qcWbuDZEkAd4rRSin5ZdsYXLcLtguGNj9HTZb`

`slot 509535911` fund oracle-3 from existing devnet wallet

`3RBGftDnSdPuxGo2anPhkUf4eRKd9CHnK4CWafHJ7aEJzjmoKu7AdnDXRmqSwhVkpsrcyjteEJz4F51EuyuywDNn`

Register every signing producer

The admin registered the three new oracle keys. The devnet admin also signs the single consumer.

slot 509535922 register 6mTM9YQcxGdkPmAzGCF25vCkdCpJqbjEYHxChvHfeTwH

4hCsf2LwFGxLLJieBMaSGihPDo911p5XAfHMeapPCLAfASC4XcUuvGkjRaXJnTG5fd4FuVVKmmjCLoWpRTXGCJpH

slot 509535932 register 7tx3XxRt1u1bfsiwrjpaJ59hbr483Fvw1YiRmvs7ohZf

3pZYk6HQwuK5iU5hpghHdUJS29FaZtRYakkWFpLwVmsLahPrV3p4pBUP1Kdk5JQkyuXwf3Xmrncx4fpZ5rQiT7TD

slot 509535940 register FLj3KsBGXCh9uM7PY3ttKz13MpLvrj98i2qWY8daaa6L

4DavqzKvEJY4uUyq2LPSnxhXynoKLcw2CYohiLmiQkjq2R45juxTjsVf5kUb6znLVrjNJ5uZnrY7EokkTGHccJQD

> Decoded OperatorRegistry contains all three new keys. Existing registered test operators remain present.

The same asset on all three venues

Venue	APY sample	Verified account
Marginfi	6.99%	2s37akK2eyBbp8DZgCm7RtsaEz8eJP3Nxd4urLHQv7yB
Save	1.64%	BgxfHJDzm44T7XG68MYKx7YisTjZu73tVovyZSjJMpmw
Kamino	4.51%	D6q6wuQSrifJKZYpR1M8R4YawnLDtDsMmWM1NbBmgJ59

Mint: `EPjFWdd5AufqSSqeM2qN1xzybapC8G4wEGGkZwyTDt1v`

Independent fetches verify mainnet genesis, account owner, mint, market and supported layout. Raw bytes and RPC context slots are retained.

Base lending APY excludes token incentives. Kamino legacy reserves use the SDK nominal 500ms slot convention.

Three APY feeds, three votes required

Venue	Pool / data	Feed PDA
Marginfi	80 / 0	A5XBGuLiiFmhn83BZXNz6kF5PR6vehVRDtaFvjq4mV65
Save	81 / 0	HXTs2z35oLW5DvUiyMuEnWo7yNXRDp3FYrG7oA22Jvtb
Kamino	82 / 0	H2WGBKwRwRxagHUvsaCUTVPBAjPisC6L9JWDV1E5gsSp

1 bps = 0.01%. Median aggregation. Minimum and maximum interval = 240 slots. Ring capacity = 256 samples.

`slot 509535960` initialize Marginfi USDC supply APY feed

65E676L5jVENaim1vzuP9tRAuhLs6CMqkaBEBghkVDpbDPj9tBu5Eo9ruh8aLG2xSm5mqZ1g5KaTNxWc3Wdhu3LK

`slot 509535972` initialize Save USDC supply APY feed

44vbrYhDYpqPPX6vGLZ89pMhjZQ4G5Qkk7SomgYBgwC63zPXnAku96gKD52Rta7GwE5GtwcCNjhMz6QooWyg9Bp7

`slot 509535978` initialize Kamino USDC supply APY feed

2YMqA3rargAYgsaukvWkqgyscGuwpQstCKDMy19w2XWhA8ComzcWiEbtPY9meyu5bUMPjsMbafYA2x8YeDcCcYvd

Four isolated processes stay running

Container	Signer	Status
oracle-1	6mTM9YQcxGdkPmAzGCF25vCkdCpJqbjEYHxChvHfeTwH	running / healthy
oracle-2	7tx3XxRt1u1bfsiwrjpaJ59hbr483Fvw1YiRmvs7ohZf	running / healthy
oracle-3	FLj3KsBGXCh9uM7PY3ttKz13MpLvrj98i2qWY8daaa6L	running / healthy
consumer	2rH2YbtCjFxNV4JL6JC1AQixxxviBEmMk8aBYBABoukc	running / healthy

Each producer receives only its own secret. The consumer receives the devnet admin secret. Services restart unless stopped and use read-only filesystems apart from mounted journals.

```
sudo docker compose -f deployment/devnet/compose.yaml ps
```

Decode the three signed votes

> Window 2123087 · closes at 509541120 · finalized · votes 3 / 3 · median 452 bps

Signing operator	Submitted APY (bps)
FLj3KsBGXCh9uM7PY3ttKz13MpLvrj98i2qWY8daaa6L	452
7tx3XxRt1u1bfsiwrjpaJ59hbr483Fvw1YiRmvs7ohZf	452
6mTM9YQcxGdkPmAzGCF25vCkdCpJqbjEYHxChvHfeTwH	452

R7ejTyc5gwVFooSBpxi9GcYogdCb4d4vTJzi3t955tn

slot 509540912 propose_sample / corroborate_sample USDC venues window=2123087

3SEM2FxmhLJs9SCzcTG6TWZmYbJoLbfqMwVqJahmGN626oZf98SMVvigE2mqJ4EgtX42nxBW6bWoJsfydy2qzTuJ

slot 509540968 propose_sample / corroborate_sample USDC venues window=2123087

5Hgiq3Jfp75s1T5r7yAbp1EcbULiwmG9BP4ac12rJ6MMNfYADrRDpYEUkS7rkkbZfVHvN9yzHhqvha3v6wQt3JbQ

The observer re-decodes archived bytes, verifies the three submitters and recomputes the median.

Yield → normalize → cap

Venue	Input APY bps	Risk	Cap	Weight
Marginfi	699	0 bps	50%	50.00%
Save	164	0 bps	50%	14.70%
Kamino	452	0 bps	50%	35.30%

Integer arithmetic; largest-remainder rounding; cap redistribution proportional to available headroom. The demo JS calculation was checked against the production Rust rubric.

> Risk discount = 0 is a neutral demonstration policy, not a protocol risk assessment.

Open, submit, finalize on chain

slot 509539012 open allocation 509539005 quorum=1

3akvoLNC4LQT1Y8m9ToodbZG4geTfC5ajVmh8YkBBixAdowTced8rtzSQhUGB4KvJCwHBNAqGJQ2HHs7cDvwzfN2

slot 509539020 submit allocation 509539005

QcaAPddUsXN4gasqjZwPAPJAtdeHcp7ZhS1Ns3BxXwxBRnuGK7h87VgDxDEccHas2iiwacRduyZ7hTpUGLRNUnW

slot 509539116 finalize allocation 509539005

3z8zSz2AS2dmdNQEB9GTvy1ozBKyZ8aJP6dZ1TKaty1keHHfhaFUGeEDkiAqTpNZa67EWqrAHKpaLwoFg8G7Si6x

The consumer starts a fresh round when all three feed heads advance together, waits for the closing slot, and finalizes the weights.

AllocationState advances over time

Fxm3c6cfmHb3taPdqVcka6USRAHkytZn3zyBUAwPbS8S



15 snapshots over 240 seconds; 4 allocations published during observation.

● Marginfi ● Save ● Kamino


```
{
  "round": "509541197",
  "updated_slot": "509541307",
  "weights": [
    5000,
    1470,
    3530,
    0,
    0,
    0,
    0,
    0,
    0,
```

Archive proof, then return rent

slot 509536852 reclaim Save sample rent 2123069

5ForkM8WXant49sVojYktsKdpq6aTPYv1pdq8hi3gdRYhwKNhNHMQqHexoqKwRW5bAmt2vSeQRfr5dJbL32xgoFv

slot 509536864 reclaim Kamino sample rent 2123069

4qYbFU8d6GNzUDSFBY43biN5id5Zpe8q1k9nTEqACPj9HtTVMSoS2GoKD9qGjMc5sGm5Y6oNuXJtLemTeHmvLHVc

Persistent ring buffers and AllocationState remain on chain. Completed consensus bytes are archived before closing. Submissions and finalization instructions remain verifiable in fetched transactions.

Devnet checks rejected open-account closure and unauthorized rent reclamation. The allocation closure test returned rent minus its transaction fee.

Verified and still running

Marginfi	Save	Kamino
50.00%	**14.70%**	**35.30%**

Weights sum to 100%. The three oracles and consumer were verified running and healthy after observation.

34 finalized venue windows verified; 203 confirmed transactions fetched and decoded.

Devnet fee funding is finite. This system publishes allocation signals; it does not move capital.

Reproduce and inspect every step

```
./scripts/devnet/deploy-system.sh
```

Outcome Raw + decoded PDAs Fetched + decoded transactions Sources + consensus archives SHA-256 manifest

Official SDKs: Marginfi · Save · Kamino

Static neutral risk discount (0 bps) and 50% per-venue caps are an explicit demo policy.
Base supply interest APY excludes token incentive emissions.
The single consumer uses the devnet registry admin to open rounds; producer containers only receive their own key.
Samples attest agreement by three operators using the same public mainnet RPC; this is not source decentralization.
Latest feed reads have an RPC context lower bound; historical input commitments are not implemented.